

LVDS Transmitter with Integer/Fractional PLL

I/O Datasheet

Chipalooza 2026 · IHP SG13CMOS5L

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1 Description

A PRBS-7 pattern generator and an LVDS output driver, clocked either directly from the reference or from an on-chip PLL whose feedback ratio is programmable over the shared digital inputs. The port list is `verilog/rtl/user_project_wrapper_4a.v` of `RTimothyEdwards/sg13cmos5l_ocd_chipalooza`, with every bus expanded into individual pins. Four dedicated analog pads means wrapper `_4a`, and therefore slot `s1` or `s16` — per `config.txt` the only two with four.

Signal flow: the reference arrives on `analog_pin[0]`; the PLL multiplies `DIV_RATIO` and halves the result; `lvds_pattern` selects reference or PLL clock, generates PRBS-7 and hands a differential pair to `lvds_tx`, which drives two of the analog pads.

2 Pin summary

All 54 ports of the wrapper. *Not connected* means the pin exists in the port list but reaches no logic inside this project.

Pin	Dir	Function
<i>Supplies</i>		
<code>vdd_3v3</code>	inout	3.3 V — driver, pre-driver, output stage
<code>vss_3v3</code>	inout	3.3 V return
<code>vdd_1v2</code>	inout	1.2 V — pattern generator, PLL
<code>vss_1v2</code>	inout	1.2 V return
<code>vssio</code>	inout	not connected
<i>Clock and enable</i>		
<code>clk</code>	in	not connected
<code>enable</code>	in	not connected — see §6
<i>Pattern generator control</i>		
<code>dig_in[0]</code>	in	<code>clk_src</code> : 0 takes the pad, 1 takes the PLL
<code>dig_in[1]</code>	in	<code>en</code> — gates the pattern clock
<code>dig_in[2]</code>	in	<code>reset</code> , active high, seeds the PRBS
Neither reaches the two output flops: they run on an ungated clock with <code>RESET_B</code> tied high, so <code>D_p</code> and <code>D_n</code> are complementary from the first edge after power-up and the driver settles once rather than at every enable.		
<code>dig_in[3]</code>	in	<code>mode</code> : 0 clock passthrough, 1 PRBS-7
<code>dig_in[4]</code>	in	not connected
<i>PLL control</i>		
<code>dig_in[5]</code>	in	PLL ENABLE
<code>dig_in[6]</code>	in	PLL RESET_N, active low
<code>dig_in[16:7]</code>	in	<code>DIV_RATIO[9:0]</code> , unsigned Q7.3 — see §4
<code>dig_in[18:17]</code>	in	<code>TEST_DIV[1:0]</code> — selects the <code>TEST_CLK</code> divider
<code>dig_in[23:19]</code>	in	not connected
<i>Digital outputs</i>		
<code>dig_out[11:0]</code>	out	not connected
<i>Dedicated analog pads</i>		
<code>analog_pin[0]</code>	inout	<code>ref_clk</code> — PLL reference, 25 to 250 MHz verified
<code>analog_pin[1]</code>	inout	<code>TEST_CLK</code> brought off chip

Pin	Dir	Function
analog_pin[2]	inout	LVDS output, non-inverting
analog_pin[3]	inout	LVDS output, inverting
<i>Bias references</i>		
ibias[0]	in	2 μ A — pre-driver, mirrored 1:15 internally
ibias[1]	in	2 μ A — driver, mirrored 1:15 internally
vbias	in	not connected
analog_bus[0]	inout	2 μ A — PLL charge pump
analog_bus[1]	inout	1.2 V — LVDS common-mode reference
analog_bus[3:2]	inout	not connected

3 Recommended operating conditions

Parameter	Pin	Min	Typ	Max	Unit
Core supply	vdd_1v2	1.08	1.20	1.32	V
Driver supply	vdd_3v3	2.97	3.30	3.63	V
Reference clock	analog_pin[0]	25	250	500	MHz
Bias current	ibias[1:0]	0.05	2	10	μ A
Charge-pump bias	analog_bus[0]	0.05	2	10	μ A
Common-mode ref.	analog_bus[1]	—	1.2	—	V

The bias limits are those of the harness current DACs, whose range is 50 nA to 10 μ A. The blocks themselves were characterised at 30 μ A; the 1:15 pre-mirrors in `lvds_tx` bridge the gap, so the pins take 2 μ A while the driver and pre-driver still see the bias they were measured at.

4 PLL configuration

The feedback ratio is one unsigned Q7.3 word on `dig_in[16:7]`: `div_ratio[9:3]` is the integer part and `div_ratio[2:0]` the eighths. `fractional_divider.v` requires the integer part to be at least 4, so the usable range is 4.000 to 127.875 in steps of 0.125.

$$f_{VCO} = f_{\text{ref}} \times \text{DIV_RATIO} \quad f_{\text{pll_clk}} = \frac{f_{VCO}}{2} \quad f_{\text{TEST_CLK}} = \frac{f_{VCO}}{2^{1+\text{TEST_DIV}}}$$

The output divider always halves, so the line rate equals $f_{\text{ref}} \times \text{DIV_RATIO}/2$. The ring oscillator covers 322 to 4050 MHz at typical / 1.2 V / 27 °C and does not start below 0.50 V of control voltage, so combinations whose VCO frequency falls outside that band will not lock.

Worked example. 250 MHz on `analog_pin[0]` and a of 500 Mbit/s wants `DIV_RATIO = 4.0`, code $4.0 \times 8 = 32 = 0000100000_2$. Only bit 5 is set, which is `dig_in[12]`; every other ratio bit is driven low. The VCO then runs at 1 GHz.

Fractional operation. The divider is a plain $N/N+1$ accumulator with no delta-sigma shaping. At a fractional ratio the instantaneous period alternates and only the average over a full accumulator cycle is the intended frequency; expect corresponding spurs on the control voltage.

5 Electrical characteristics

Simulated at the top level, typical corner, 27 °C, over 6 μ s, with the reference at 250 MHz and `DIV_RATIO = 4.0`. Load is $49.9 \Omega + 49.9 \Omega$ across the two output pads with the centre tap giving V_{OS} . **No pad or ESD model is included**, so V_{OS} here is not the compliance figure.

Parameter	Symbol	Measured	Limit	Reference
Differential output	$ V_{OD} $	380.3	247 ... 454	TIA/EIA-644-A 4.1.1
Offset voltage	V_{OS}	1193.9	1125 ... 1375	4.1.2
Offset, peak-to-peak	$V_{OS,pp}$	118.1	< 150	4.1.5
Offset p-p, first 100 ns		159.4		cold start, see §6
PLL output frequency	f_{pll_clk}	500.0117	500	target
Frequency error		23.3		ppm
Control voltage	V_{CTRL}	581.4		mV, settled
Driver supply current	I_{3v3}	5.74		mA
Core supply current	I_{1v2}	0.341		mA

Voltages in mV unless noted. The frequency and control-voltage figures come from the co-simulated PLL, in which the phase detector and both dividers are the RTL of `macros/pll_digital` executed through ngspice's `d_cosim`. Supply currents are from the LVDS bench, which clocks the pattern generator from the reference directly.

6 Known limitations

- **No ESD structures or pad models.** Clause 4.1.4 and 4.1.5 figures depend on pad capacitance, which is not represented here.
- **enable has no load.** The pattern clock is gated by `dig_in[1]` alone. The harness masks `dig_in` to zero for an unselected project, so an unselected project is still idle — but `select` and `dig_ena` high with `enable` low leaves the block running with the project enable deasserted.
- **ΔV_{OS} is not measured at the top level.** The 50 mV limit of clause 4.1.2 is checked only in the macro's own DC bench.
- **Offset margin is asymmetric, and the reference is unsettled.** The benches drive `analog_bus[1]` at 1.2 V, where V_{OS} sits 69 mV above the lower limit of clause 4.1.2 and 181 mV below the upper. The rest of the documentation calls this reference 1.25 V, which is where the clause centres and would split the margin evenly. Which of the two the harness bias generator can actually deliver is unverified, so the figures here are the 1.2 V ones and the discrepancy is open.
- **$V_{OS,pp}$ is switching feedthrough, not settling.** It is a 93 mV dip at every single data edge, and it starts in the pre-driver rather than the driver. The falling gate edge is 103.5 ps against 112.0 ps rising, so at each crossing the gate common mode $(V_{Inp} + V_{Inn})/2$ dips 78 mV; that lands on the output common mode at a ratio of 1.11, identically in both data directions. The cross-coupled neutralisation caps cannot cancel it — they remove differential feedthrough and this term is common mode — and the CMFB loop sees only 36.8 mV of it through its sense filter, far too slow for a 165 ps event in any case. The tail current imbalance was measured and ruled out: 3.3 fC where 34.6 fC would be needed. Two levers are characterised but not adopted — the pre-driver's last PMOS has a narrow optimum at 36 to 38 μm giving 102 mV, and a common-mode RC damper across the outputs gives 97 mV for 2.6 % of $|V_{OD}|$.
- **The first 100 ns are outside the compliance window.** $V_{OS,pp}$ reaches 159.4 mV there while the driver's common-mode loop settles from the DC operating point. The benches measure the compliance figures from 150 ns onwards. This is what the ungated, reset-free output flops improved: the same window measured 288.3 mV when the pair was not complementary until the pattern was enabled.
- **`analog_pin[1]` is driven by TEST_CLK with no pad driver in front of it.**
- **`core_p` / `core_n` cross from `vss_1v2` into the pre-driver on `vss_3v3`;** the two returns have to be tied in the frame.
- **Simulation needs a build step.** The PLL's digital half runs as a Verilator-compiled shared object under `d_cosim`. Run `make pll-cosim-so` before any top-level simulation; the Makefile and the Simulate arrow in xschem both do this automatically.